

TECHNICAL REPORT

AI-TOD vs VisDrone

Dataset Comparison Report

Graphical & Mathematical Analysis · Object Size Distribution · RT-DETR Query Saturation · NWD Loss Theory · Deployment Recommendations

|                        |             |                          |                  |
|------------------------|-------------|--------------------------|------------------|
| AI-TOD Images          | 28,036      | VisDrone Images          | 10,209           |
| AI-TOD Avg Object Size | ~12.8 px    | VisDrone Avg Object Size | ~40+ px          |
| AI-TOD < 32 px objects | 98.3%       | VisDrone < 32 px objects | 74.5%            |
| AI-TOD < 16 px objects | ~86%        | VisDrone < 16 px objects | ~22%             |
| AI-TOD max objects/img | 900+        | VisDrone typical objects | 20–200           |
| RT-DETR Q (default)    | 300 queries | NWD Constant C           | 12.8 px (AI-TOD) |

1. INTRODUCTION

Object detection from aerial imagery has become a critical research frontier in computer vision, with applications spanning UAV-based traffic monitoring, satellite intelligence, disaster response, border surveillance, and autonomous drone operations. Among the most widely studied benchmarks for aerial object detection are the AI-TOD (Artificial Intelligence Tiny Object Detection) dataset and the VisDrone dataset — two collections that, despite sharing an aerial perspective, present fundamentally different challenges to modern detectors.

AI-TOD focuses on satellite and high-altitude UAV imagery where objects of interest are extremely small — ships, aircraft, vehicles, and infrastructure features that may occupy fewer than 16 pixels in the source image. VisDrone, by contrast, captures lower-altitude drone footage of urban environments, where vehicles and pedestrians are more visually prominent but appear in densely packed, occluded scenes.

These differences have direct and profound consequences for the design of transformer-based detectors such as RT-DETR. In particular, the number of learnable object queries — the fixed-size set of prediction slots in the transformer decoder — must be carefully calibrated to the maximum expected object density in each dataset. Failure to do so creates a hard architectural ceiling on recall that no amount of training can overcome.

This report provides a structured, graphical, and mathematical analysis of both datasets, quantifies the RT-DETR query saturation problem specific to each, derives the formal conditions under which query bottlenecks occur, and concludes with concrete deployment recommendations tailored to each benchmark.

2. DATASET OVERVIEW & PROPERTIES

The table below summarises the key structural properties of both datasets that govern detector design decisions:

| Property                | AI-TOD                                    | VisDrone                                         |
|-------------------------|-------------------------------------------|--------------------------------------------------|
| Domain                  | Remote sensing (satellite / high-alt UAV) | UAV urban scenes (low-altitude drone)            |
| Total images            | 28,036                                    | 10,209                                           |
| Object classes          | 8                                         | 10                                               |
| Average object size     | ~12.8 pixels                              | ~40+ pixels                                      |
| Objects < 32 px         | 98.3% — nearly the entire dataset         | 74.5% — substantial but with larger tail         |
| Objects < 16 px         | ~86% — dominant challenge                 | ~22% — smaller proportion                        |
| Typical objects         | Ships, aircraft, vehicles, wind-mills     | Cars, pedestrians, bicycles, trucks, bus         |
| Scene type              | Top-down satellite / high-altitude view   | Oblique / overhead low-altitude urban view       |
| Objects per image       | Up to 900+ (dense port / city scenes)     | Typically 20–200                                 |
| Main challenge          | Sub-pixel tiny objects; query saturation  | Crowded urban scenes; occlusion; class diversity |
| NWD constant C          | 12.8 px (matches avg object scale)        | ~16 px (larger objects warrant larger C)         |
| Recommended Q (RT-DETR) | 500–900 (vs default 300)                  | 300 (default sufficient)                         |

Dataset Scale Asymmetry



AI-TOD contains nearly 3x as many images as VisDrone (28,036 vs 10,209) and up to 4.5x more objects per image in dense scenes. This combination makes AI-TOD both the larger training set and the more demanding inference-time challenge.

### 3. OBJECT SIZE DISTRIBUTION ANALYSIS

Object size is the single most impactful property differentiating these two datasets. It governs feature extraction quality, regression loss effectiveness, and the need for super-resolution preprocessing. The following analysis breaks down size distribution across three critical thresholds.

#### 3.1 Average Object Size — Implications for Backbone Feature Maps

The average object size in AI-TOD is approximately 12.8 pixels, compared to 40+ pixels in VisDrone. This disparity has compounding consequences as the detection backbone progressively downsamples spatial resolution:

| Dataset  | Avg Object Size | At Stride 8 (P3)    | At Stride 16 (P4)    | At Stride 32 (P5)   | Visibility          |
|----------|-----------------|---------------------|----------------------|---------------------|---------------------|
| AI-TOD   | 12.8 px         | 1.6 px — borderline | 0.8 px — sub-pixel   | 0.4 px — invisible  | Critically limited  |
| VisDrone | ~40 px          | 5.0 px — detectable | 2.5 px — challenging | 1.25 px — difficult | Manageable with FPN |
| VisDrone | ~100 px         | 12.5 px — good      | 6.25 px — detectable | 3.1 px — acceptable | Fully visible       |

At the P5 feature level (stride 32), an AI-TOD object averaging 12.8 px projects to a 0.4 px feature — effectively a phantom. Even at P3 (stride 8), the 1.6 px projection is at the boundary of reliable detection. This is why AI-TOD mandates special-purpose techniques that standard aerial datasets do not require: super-resolution preprocessing to inflate effective object size, SAHI tiling to reduce the spatial scale at inference, and NWD loss to provide gradients when bounding boxes do not overlap.

#### 3.2 Proportion of Objects Below 32 px (COCO 'Small' Threshold)

The COCO benchmark defines objects smaller than 32×32 pixels as 'small'. The following visual comparison illustrates how both datasets relate to this threshold:

| Category | Visual      | Value / % |
|----------|-------------|-----------|
| AI-TOD   | <div></div> | 98.3%     |
| VisDrone | <div></div> | 74.5%     |

AI-TOD is almost entirely composed of small objects (98.3%), making the COCO definition of 'small' essentially the default category rather than the exceptional case. VisDrone's 74.5% is still a majority but includes a meaningful tail of medium and large objects that standard detection pathways can handle without specialised treatment.

#### 3.3 Proportion of Objects Below 16 px (Sub-Feature Level)

Objects below 16 pixels are particularly challenging because they fall below the feature resolution at stride 16 (P4), a commonly used detection scale:

| Category | Visual                 | Value / % |
|----------|------------------------|-----------|
| AI-TOD   | <div><div></div></div> | 86%       |
| VisDrone | <div><div></div></div> | 22%       |

AI-TOD has approximately 86% of objects below 16 pixels — nearly four times the proportion in VisDrone (~22%). This 4x gap directly motivates the mandatory use of Real-ESRGAN super-resolution for AI-TOD: a 2× upscale converts a 12.8 px average object to 25.6 px, bringing it meaningfully above the P4 feature floor and dramatically improving the detectability of the sub-16 px majority.

3.4 Object Size Regime Summary

| Size Regime                      | AI-TOD Proportion | VisDrone Proportion | Detection Treatment                    |
|----------------------------------|-------------------|---------------------|----------------------------------------|
| < 8 px (invisible at stride 8)   | ~40%              | ~5%                 | ESRGAN 4× + SAHI essential             |
| 8–16 px (borderline at stride 8) | ~46%              | ~17%                | ESRGAN 2× strongly recommended         |
| 16–32 px (COCO small)            | ~12.3%            | ~52%                | SAHI tiling helps; NWD loss beneficial |
| 32–96 px (COCO medium)           | ~1.5%             | ~18%                | Standard FPN detection path            |
| > 96 px (COCO large)             | ~0.2%             | ~8%                 | Standard detection, high confidence    |

4. RT-DETR QUERY SATURATION — MATHEMATICAL ANALYSIS

The query saturation problem is unique to DETR-family detectors and represents the most critical architectural challenge specific to high-density datasets like AI-TOD. Understanding it rigorously is essential for any practitioner deploying RT-DETR on aerial imagery.

4.1 The Object Query Mechanism

RT-DETR and all DETR-family detectors use a fixed set of  $Q$  learnable object queries — embedding vectors that are processed by the transformer decoder to produce exactly one predicted bounding box and class score each. During training, each ground-truth object is matched to exactly one query via the Hungarian bipartite matching algorithm. This creates a hard structural constraint:

$Q \geq N$  (required condition for zero structural recall loss)

When the number of ground-truth objects  $N$  in an image exceeds the query count  $Q$ , the Hungarian matcher cannot assign a unique slot to every object. At least  $N - Q$  objects will have no associated query prediction, creating an inherent recall ceiling that is independent of model capacity, training duration, or optimisation quality.

Structural Recall Ceiling Formula

X

Maximum Recall  $\leq Q / N$  when  $N > Q$ . This is not a training failure — it is a fundamental architectural constraint. A model with  $Q=300$  queries on an image with  $N=900$  objects can achieve at most 33.3% recall regardless of any other factor.

4.2 Query Utilisation Ratio — Formal Definition

Define the Query Utilisation Ratio  $R$  as the ratio of objects to available prediction slots:

$R = N / Q$

| Condition          | Regime             | Structural Impact                               | Required Action                         |
|--------------------|--------------------|-------------------------------------------------|-----------------------------------------|
| $R < 0.5$          | Comfortable        | Large headroom — no concern                     | None                                    |
| $0.5 \leq R < 0.8$ | Approaching        | Reasonable margin — monitor dense scenes        | Consider increasing $Q$ if $N$ grows    |
| $0.8 \leq R < 1.0$ | Critical margin    | Near-limit — any extra object causes saturation | Increase $Q$ pre-emptively              |
| $R = 1.0$          | Perfect fit        | Zero slack — one extra object → loss            | Increase $Q$ by 20% safety buffer       |
| $R > 1.0$          | Saturated          | Hard recall ceiling at $Q/N$                    | Must increase $Q$ to at least $N_{max}$ |
| $R > 2.0$          | Severely saturated | Most objects undetectable                       | SAHI tiling + $Q$ increase essential    |

### 4.3 VisDrone — No Bottleneck ( $R < 1$ )

For a typical VisDrone urban traffic scene with moderate density:

| Parameter                         | Value             | Interpretation                             |
|-----------------------------------|-------------------|--------------------------------------------|
| N — objects per image (typical)   | ~120              | Cars, pedestrians, cyclists in urban frame |
| N — objects per image (dense)     | ~200              | Busy intersection or market scene          |
| Q — RT-DETR-L default query count | 300               | Standard configuration                     |
| R (typical)                       | $120/300 = 0.40$  | 40% utilisation — large headroom           |
| R (dense)                         | $200/300 = 0.67$  | 67% utilisation — approaching but safe     |
| Unassigned queries (typical)      | $300 - 120 = 180$ | 180 free prediction slots                  |
| Maximum structural recall loss    | Zero              | All objects can be assigned a query        |
| Status                            | No bottleneck     | Default Q=300 is fully adequate            |

$$R_{\text{VisDrone}} = 120 / 300 = 0.40 \rightarrow R < 1 \quad (\text{no structural bottleneck})$$

#### VisDrone: Default Configuration Sufficient



RT-DETR-L with Q=300 queries is fully adequate for VisDrone across all typical scene densities. Even in the densest urban scenes (N~200), the utilisation ratio R=0.67 provides a comfortable safety margin. The primary detection challenges for VisDrone are occlusion, class similarity, and viewpoint variation — not query count.

### 4.4 AI-TOD — Critical Query Saturation ( $R \gg 1$ )

For a dense AI-TOD satellite image of a port, airport, or urban intersection:

| Parameter                       | Value             | Interpretation                                        |
|---------------------------------|-------------------|-------------------------------------------------------|
| N — objects per image (typical) | ~400              | Ships, vehicles, aircraft in satellite image          |
| N — objects per image (dense)   | ~900              | Crowded port or urban cluster                         |
| N — objects per image (extreme) | ~1500             | AI-TOD-V2 extreme density scenes                      |
| Q — RT-DETR-L default           | 300               | Standard configuration                                |
| R (typical)                     | $400/300 = 1.33$  | 33% structural recall loss — 100 objects undetectable |
| R (dense)                       | $900/300 = 3.00$  | Only 1/3 of objects can be assigned a query           |
| R (extreme)                     | $1500/300 = 5.00$ | 80% of objects structurally undetectable              |
| Objects undetectable (dense)    | $900 - 300 = 600$ | 600 objects guaranteed to be missed                   |
| Status                          | QUERY SATURATION  | Q must be increased significantly                     |

$$R_{AITOD(dense)} = 900 / 300 = 3.0 \rightarrow R > 1 \text{ (severe saturation)}$$
$$\text{Max Recall(dense)} = Q / N = 300 / 900 = 33.3\% \text{ (structural ceiling)}$$

**AI-TOD Query Saturation: A Structural Recall Ceiling**



With  $R = 3.0$ , the model is structurally limited to detecting at most 33.3% of objects in a dense AI-TOD scene — 600 out of 900 objects physically cannot be assigned a prediction query. This is not caused by insufficient training, poor learning rate, or weak architecture. It is a consequence of  $Q < N$  and cannot be fixed by any training procedure. The minimum fix is  $Q \geq N_{max}$  for the dataset.

4.5 Visual Comparison of Query Capacity Utilisation

| Category                 | Visual                 | Value / % |
|--------------------------|------------------------|-----------|
| VisDrone Sparse (N=30)   | <div><div></div></div> | 10%       |
| VisDrone Typical (N=120) | <div><div></div></div> | 40%       |
| VisDrone Dense (N=200)   | <div><div></div></div> | 67%       |
| AI-TOD Typical (N=400)   | <div><div></div></div> | 100%      |
| AI-TOD Dense (N=900)     | <div><div></div></div> | 100%      |

Note: AI-TOD Typical ( $R=1.33$ ) and AI-TOD Dense ( $R=3.0$ ) are both shown at 100% because they both exceed  $Q$  — the bars represent the capacity ceiling being hit, not the actual utilisation ratio (which would be 133% and 300% respectively if the chart extended that far).

5. LOCALISATION ERROR — WHY AI-TOD IS MATHEMATICALLY HARDER

Even when a tiny object successfully receives a query assignment, accurate bounding box regression presents a second, distinct challenge. The same absolute pixel error in box coordinates has a profoundly different impact on detection quality depending on the object size.

5.1 Relative Localisation Error Formula

Define relative localisation error as the ratio of absolute prediction error to the object's true spatial extent:

Relative Error = Δx / w

where Δx is the absolute pixel error in box center prediction and w is the ground-truth object width in pixels. For a typical detector with a realistic localisation noise of 2 pixels:

| Dataset  | Object Width (w) | Absolute Error (Δx) | Relative Error (Δx/w) | Quality Impact                            |
|----------|------------------|---------------------|-----------------------|-------------------------------------------|
| AI-TOD   | 8 px             | 2 px                | 25.0%                 | Box severely misaligned — likely mismatch |
| AI-TOD   | 12 px            | 2 px                | 16.7%                 | Box significantly offset — poor IoU       |
| AI-TOD   | 16 px            | 2 px                | 12.5%                 | Box noticeably offset — borderline        |
| VisDrone | 40 px            | 2 px                | 5.0%                  | Box mildly offset — usable                |
| VisDrone | 80 px            | 2 px                | 2.5%                  | Box accurate — good IoU                   |
| VisDrone | 120 px           | 2 px                | 1.7%                  | Box highly accurate — excellent IoU       |

Relative Error Ratio = 16.7% / 5.0% = 3.34× (AI-TOD worse than VisDrone)

The same 2-pixel prediction error is 3.34× more damaging for AI-TOD than for VisDrone in terms of relative box accuracy. Standard regression losses (L1, Smooth-L1, GloU) treat this 2-pixel error identically regardless of object size — they are scale-blind and therefore inherently biased against tiny objects.

5.2 IoU Behaviour for Tiny Objects

The Intersection-over-Union metric exhibits a particularly pathological behaviour at small scales. For two boxes of size w × h displaced by d pixels with zero overlap:

IoU = 0 when d > 0 (any non-zero displacement → zero overlap)



$$\partial(\text{IoU}) / \partial(\text{prediction}) = 0 \quad (\text{zero gradient} - \text{learning stops})$$

For an AI-TOD object of 8×8 pixels, any displacement of more than 8 pixels in any direction produces IoU = 0. For a VisDrone car of 40×30 pixels, IoU remains non-zero for displacements up to ~40 pixels. The region of informative IoU gradient is proportional to object area — 25× larger for VisDrone than for the smallest AI-TOD objects.

### 5.3 NWD Loss — Scale-Aware Gradient Signal

The Normalized Wasserstein Distance loss addresses both the relative error problem and the zero-IoU gradient problem simultaneously by modelling each bounding box as a 2D Gaussian distribution:

$$\begin{aligned} \text{Box } G &\rightarrow \text{Gaussian } N(\mu, \Sigma) \quad \text{where } \mu = (cx, cy), \quad \sigma = (w/2, h/2) \\ W_2(G_1, G_2) &= \sqrt{\|\mu_1 - \mu_2\|^2 + \|\sigma_1 - \sigma_2\|^2} \\ \text{NWD}(G_1, G_2) &= \exp(-W_2(G_1, G_2) / C) \end{aligned}$$

The critical property is that  $\sigma = (w/2, h/2)$  normalises the distance by object size. A 2-pixel displacement on an 8-pixel object and a 2-pixel displacement on a 40-pixel object produce different NWD values — the tiny object receives a larger penalty, correctly reflecting its higher relative degradation. NWD also remains strictly positive ( $> 0$ ) for any finite displacement, providing non-zero gradients even when IoU = 0.

| Loss Function | Non-Overlapping Gradient | Scale Awareness                   | Tiny Object Sensitivity | Best For              |
|---------------|--------------------------|-----------------------------------|-------------------------|-----------------------|
| IoU           | 0 — learning stops       | None — absolute pixel scale       | Very Low                | Large objects only    |
| GIoU          | Partial — some signal    | None — absolute pixel scale       | Low                     | Medium/large objects  |
| DIoU          | Partial                  | None                              | Low                     | Medium objects        |
| NWD           | Always $> 0$             | Full — normalised by $\sigma=w/2$ | High                    | Tiny objects (AI-TOD) |
| NWD+IoU       | Always $> 0$             | Partial                           | High                    | Mixed datasets        |

#### ★ NWD Constant C — Physical Interpretation

The constant C in  $\text{NWD} = \exp(-W_2/C)$  normalises the Wasserstein distance to the mean object scale of the dataset. For AI-TOD,  $C = 12.8$  px equals the mean object size, ensuring that an object displaced by one average object-width receives  $\text{NWD} \approx \exp(-1) \approx 0.37$  — a meaningful but not catastrophic penalty. For VisDrone with larger objects, C should be increased to ~16 px to maintain the same semantic interpretation.

## 6. RECOMMENDED RT-DETR CONFIGURATIONS

Based on the mathematical analysis in Sections 4 and 5, the following configurations are recommended for each dataset and density regime:

### 6.1 Query Count (Q) Recommendations

| Dataset  | Density Regime | Typical N | Recommended Q | R at Recommended Q | Rationale                              |
|----------|----------------|-----------|---------------|--------------------|----------------------------------------|
| VisDrone | Sparse         | < 50      | 300           | 0.17               | Default — very comfortable margin      |
| VisDrone | Typical        | < 150     | 300           | 0.50               | Default — sufficient headroom          |
| VisDrone | Dense          | < 300     | 300–400       | 0.75               | Buffer for occasional dense frames     |
| AI-TOD   | Typical        | < 500     | 500–700       | 0.85               | $R < 1$ — minimum viable configuration |
| AI-TOD   | Dense          | < 900     | 900           | 1.00               | Matches worst-case scene density       |
| AI-TOD   | Extreme        | < 1500    | 1000–1500     | 1.00               | For AI-TOD-V2 / extreme density        |

### 6.2 Computational Cost of Increasing Q

Decoder complexity in RT-DETR scales as  $O(Q \times D \times d^2)$  where  $D$  is the number of decoder layers and  $d$  is the feature dimension. The relative decoder cost of increasing  $Q$  from 300:

| Q    | Relative Decoder Cost | Max Objects Detectable | Memory Impact | Suitable Deployment                |
|------|-----------------------|------------------------|---------------|------------------------------------|
| 300  | 1.0× (baseline)       | 300                    | Baseline      | VisDrone all, AI-TOD sparse scenes |
| 500  | 1.67×                 | 500                    | +67% decoder  | AI-TOD typical ( $N < 500$ )       |
| 700  | 2.33×                 | 700                    | +133% decoder | AI-TOD dense validation scenes     |
| 900  | 3.0×                  | 900                    | +200% decoder | AI-TOD worst-case dense scenes     |
| 1500 | 5.0×                  | 1500                   | +400% decoder | AI-TOD-V2 extreme density          |

The practical minimum for AI-TOD deployment is  $Q = 500$ , which provides  $R < 1$  for all typical scenes ( $N < 500$ ) at a 67% decoder overhead increase. For maximum coverage including dense scenes,  $Q = 900$  is required but triples decoder compute. The SAHI tiling strategy (below) reduces per-tile  $N$  and allows  $Q = 300$  to remain viable by limiting object density per inference call.

### 6.3 Full Configuration Matrix

| Dataset         | Q   | ESRGAN SR | SAHI Tiling | NWD Loss | Severity |
|-----------------|-----|-----------|-------------|----------|----------|
| VisDrone Sparse | 300 | Optional  | Not needed  | Optional | Low      |

| Dataset          | Q    | ESRGAN SR      | SAHI Tiling        | NWD Loss    | Severity |
|------------------|------|----------------|--------------------|-------------|----------|
| VisDrone Typical | 300  | Recommended    | Recommended        | Recommended | Moderate |
| VisDrone Dense   | 400  | Recommended    | Strongly rec.      | Recommended | Moderate |
| AI-TOD Typical   | 500  | Essential (2×) | Essential          | Essential   | High     |
| AI-TOD Dense     | 900  | Essential (2×) | Essential          | Essential   | Critical |
| AI-TOD Extreme   | 1500 | Essential (4×) | Essential + 2-pass | Essential   | Critical |

## 7. SAHI TILING — REDUCING EFFECTIVE N PER INFERENCE

Slicing-Aided Hyper Inference (SAHI) is a technique that addresses both the query saturation problem and the sub-pixel object problem simultaneously by dividing each high-resolution image into overlapping tiles before detection.

### 7.1 How SAHI Reduces N Per Tile

Instead of running detection on the full image with N objects, SAHI divides it into K tiles of size  $T \times T$  with overlap O:

$$N_{\text{per\_tile}} \approx N_{\text{total}} \times (T \times T) / (I_H \times I_W \times (1 - O)^2)$$

For a 2000×2000 px AI-TOD image with N = 600 objects, sliced into 640×640 tiles with 20% overlap:

| Parameter                    | Value          | Effect                                |
|------------------------------|----------------|---------------------------------------|
| Original image resolution    | 2000 × 2000 px | Full resolution — N = 600             |
| Tile size                    | 640 × 640 px   | Sub-region for detection              |
| Overlap                      | 20% (128 px)   | Prevents border object truncation     |
| Number of tiles              | ~12 tiles      | 3×4 grid with overlap                 |
| Approximate N per tile       | ~50            | 600 / 12 = 50 objects per tile        |
| R with Q=300                 | 50/300 = 0.17  | No saturation — enormous headroom     |
| Effective object size uplift | 2.0–3.0×       | Pixels per object larger in tile view |

By reducing N per tile from 600 to ~50, SAHI transforms a severely saturated scenario (R = 2.0) into a comfortable one (R = 0.17) with the default Q = 300 configuration — without any architectural changes. The additional benefit is that each object now occupies proportionally more pixels in the tile view, directly reducing the sub-pixel problem.

### 7.2 SAHI Parameters for Each Dataset

| Dataset  | Tile Size  | Overlap      | Rationale                                                        |
|----------|------------|--------------|------------------------------------------------------------------|
| AI-TOD   | 640 × 640  | 20% (128 px) | Matches RT-DETR training resolution; reduces N to ~50 per tile   |
| AI-TOD   | 480 × 480  | 20% (96 px)  | Smaller tiles for extreme density — more tiles, lower N per tile |
| VisDrone | 640 × 640  | 10–15%       | Standard — moderate overlap sufficient for VisDrone spacing      |
| VisDrone | Full image | None         | Acceptable for sparse and typical scenes without SAHI overhead   |

## 8. COMPREHENSIVE CHALLENGE COMPARISON

| Challenge                 | AI-TOD Status                        | VisDrone Status                     | Severity                   |
|---------------------------|--------------------------------------|-------------------------------------|----------------------------|
| Object size (avg)         | ~12.8 px — sub-pixel at stride 32    | ~40 px — manageable with FPN        | <b>AI-TOD: Critical</b>    |
| Query saturation          | R = 3.0 at dense scenes              | R = 0.40 typical — no concern       | <b>AI-TOD: Critical</b>    |
| Relative localisation err | 3.34× worse than VisDrone            | Moderate — acceptable with IoU      | <b>AI-TOD: High</b>        |
| Sub-pixel objects         | ~40% below 8 px — backbone invisible | ~5% below 8 px                      | <b>AI-TOD: High</b>        |
| IoU gradient failure      | IoU = 0 for most box pairs           | IoU informative — adequate          | <b>AI-TOD: High</b>        |
| NWD loss necessity        | Essential — IoU fails at this scale  | Recommended improvement             | <b>AI-TOD: Essential</b>   |
| ESRGAN SR preprocessing   | Critical — 2× upscale mandatory      | Useful but not required             | <b>AI-TOD: Critical</b>    |
| SAHI tiling               | Essential — reduces N per tile       | Recommended for dense scenes        | <b>AI-TOD: Critical</b>    |
| Scene occlusion           | Low — top-down view, sparse spacing  | High — cluttered urban, overlap     | <b>VisDrone : Moderate</b> |
| Class diversity           | 8 classes — moderate diversity       | 10 classes — pedestrian/vehicle mix | <b>Both: Moderate</b>      |
| Annotation density        | 400–900 per image (extreme)          | 20–200 per image (moderate)         | <b>AI-TOD: High</b>        |
| Dataset size for training | 28,036 images — large corpus         | 10,209 images — smaller corpus      | <b>VisDrone : Limited</b>  |

## 9. PRACTICAL DEPLOYMENT GUIDE

### 9.1 AI-TOD Deployment Checklist

- **Query count:** Set  $Q \geq 500$  for typical scenes,  $Q = 900$  for dense scenes. Do not use default  $Q = 300$ .
- **Super-resolution:** Apply Real-ESRGAN 2× as preprocessing before detection; 4× for scenes with majority sub-8 px objects.
- **Tiling:** Use SAHI tiling with 640×640 tiles and 20% overlap to reduce effective N per tile to  $< 50$ .
- **NWD loss:** Replace GloU/IoU regression loss with NWD ( $C=12.8$ ) for the bounding box regression head.
- **Confidence threshold:** Lower confidence threshold to 0.15–0.20 during evaluation; tiny objects produce lower but valid confidence scores.
- **Training monitoring:** Monitor per-epoch query utilisation ( $N_{\text{avg}} / Q$ ) during training; if consistently  $> 0.8$ , increase  $Q$ .

### 9.2 VisDrone Deployment Checklist

- **Query count:** Default  $Q = 300$  is sufficient for all typical VisDrone densities. No change required.
- **Super-resolution:** ESRGAN 2× preprocessing is recommended for improved recall of sub-16 px objects (~22% of dataset).
- **Tiling:** Use SAHI tiling for evaluation on dense urban scenes; optional for typical traffic footage.
- **NWD loss:** NWD loss provides a moderate improvement over GloU; beneficial but not structurally required.
- **Confidence threshold:** Confidence threshold 0.25 (standard) is appropriate; increase to 0.30 for vehicle-only applications.
- **Class filtering:** Apply class filtering (classes=[2,3,5,7] for car/motorcycle/bus/truck) to suppress non-vehicle COCO detections.

### 9.3 When to Use Each Technique

| Technique                 | Use for AI-TOD                 | Use for VisDrone                | Key Benefit                                 |
|---------------------------|--------------------------------|---------------------------------|---------------------------------------------|
| Increase $Q > 300$        | Essential — $Q \geq 500$       | Not needed — $Q=300$ sufficient | Removes structural recall ceiling           |
| ESRGAN 2× SR              | Essential                      | Recommended                     | Inflates object size above P3 floor         |
| ESRGAN 4× SR              | For $< 8$ px majority scenes   | Rarely needed                   | Maximum tiny object enlargement             |
| SAHI tiling               | Essential                      | Recommended                     | Reduces N per call; boosts pixel resolution |
| NWD loss                  | Essential ( $C=12.8$ )         | Recommended ( $C=16$ )          | Scale-aware gradient for tiny boxes         |
| Hungarian matching warmup | Strongly recommended           | Optional                        | Stabilises early query assignments          |
| Lower conf threshold      | 0.15–0.20                      | 0.25 standard                   | Recovers low-conf tiny detections           |
| Class filtering           | Not usually needed (8 classes) | Recommended for vehicle-only    | Removes non-vehicle COCO detections         |

## 10. CONCLUSIONS

The following conclusions are supported by the mathematical analysis and empirical dataset statistics presented in this report:

1. **1.** AI-TOD is an extreme tiny-object benchmark where 98.3% of objects are smaller than 32 pixels and 86% are smaller than 16 pixels — making it one of the most challenging aerial detection datasets in the field. Its average object size of 12.8 px falls below the P3 feature level at stride 8, requiring super-resolution preprocessing as a mandatory step rather than an optional enhancement.
2. **2.** VisDrone focuses on urban UAV imagery with objects averaging 40+ pixels — significantly larger than AI-TOD. While 74.5% of objects are still below 32 px, a meaningful tail of larger objects allows standard detection pathways to contribute, and ESRGAN / SAHI are recommended rather than mandatory.
3. **3.** The default RT-DETR query count of  $Q = 300$  is fully adequate for VisDrone. The query utilisation ratio  $R = N/Q$  typically reaches 0.40 for average scenes and 0.67 for dense scenes — both well below the saturation threshold of  $R = 1$ .
4. **4.** AI-TOD frequently suffers from severe query saturation. Dense scenes with  $N = 900$  objects produce  $R = 3.0$ , creating a structural recall ceiling of 33.3% — meaning two-thirds of objects are physically undetectable regardless of training quality. The minimum fix is  $Q \geq N_{\text{max}}$ , requiring  $Q = 500\text{--}900$  for AI-TOD deployment.
5. **5.** The relative localisation error for AI-TOD is  $3.34\times$  worse than VisDrone for the same absolute pixel displacement. This scale asymmetry makes standard IoU-based regression losses inadequate:  $\text{IoU} = 0$  for any non-overlapping tiny boxes, halting gradient flow. NWD loss directly solves this by normalising displacement by object size ( $\sigma = w/2$ ).
6. **6.** The NWD constant  $C = 12.8$  for AI-TOD is not arbitrary — it equals the mean object scale of the dataset, ensuring that an object displaced by one average object-width receives a loss penalty of  $\exp(-1) \approx 0.37$ . For VisDrone,  $C$  should be tuned to approximately 16 px to match its larger average object scale.
7. **7.** SAHI tiling is the most computationally efficient intervention for both the query saturation and sub-pixel problems: by slicing a  $2000\times 2000$  px image into  $640\times 640$  tiles,  $N$  per tile drops from 600 to  $\sim 50$  and  $R$  drops from 2.0 to 0.17 — eliminating saturation without any architectural changes.
8. **8.** The practical recommendation is: for VisDrone, use default  $Q=300$  with ESRGAN  $2\times$  and optional SAHI; for AI-TOD, use  $Q=500\text{--}900$  with mandatory ESRGAN  $2\times$ , mandatory SAHI tiling, NWD loss ( $C=12.8$ ), and a lowered confidence threshold of 0.15–0.20.

### Core Insight

The two datasets represent opposite ends of the aerial detection challenge spectrum: AI-TOD demands architectural modifications ( $Q$  increase) and preprocessing pipelines (ESRGAN + SAHI + NWD) because standard detector assumptions break down entirely at its object scales. VisDrone's challenges are more contextual — occlusion, scene complexity, class diversity — and are addressable within the standard detection framework with moderate enhancements. Both datasets benefit from the joint ESRGAN + RT-DETR + NWD training paradigm, but AI-TOD derives structural necessity while VisDrone derives incremental improvement.





## Appendix A — Mathematical Reference

### Query Utilisation Ratio

$$R = N / Q$$

$$\text{Max Recall} \leq \min(1, Q / N) \quad \text{when } N > Q$$

### Relative Localisation Error

$$\varepsilon_{\text{rel}} = \Delta x / w$$

$$\varepsilon_{\text{rel}}(\text{AI-TOD}) / \varepsilon_{\text{rel}}(\text{VisDrone}) = (\Delta x / 12) / (\Delta x / 40) = 40 / 12 \approx 3.34\times$$

### NWD Loss

$$\text{NWD}(G_1, G_2) = \exp(-W_2(G_1, G_2) / C)$$

$$W_2(G_1, G_2) = \sqrt{(\|\mu_1 - \mu_2\|^2 + \|\sigma_1 - \sigma_2\|^2)}$$

where  $\mu = (c_x, c_y)$ ,  $\sigma = (w/2, h/2)$

$$C_{\text{AITOD}} = 12.8 \text{ px}, \quad C_{\text{VisDrone}} \approx 16 \text{ px}$$

### SAHI Tile Object Count Estimation

$$N_{\text{per\_tile}} \approx N_{\text{total}} \times (T^2) / (I_H \times I_W \times (1 - 0)^2)$$

$$R_{\text{post\_SAHI}} = N_{\text{per\_tile}} / Q \ll N_{\text{total}} / Q$$

### Decoder Complexity Scaling

$$\text{Cost}(Q) = O(Q \times D \times d^2)$$

$$\text{Cost}(Q=900) / \text{Cost}(Q=300) = 3.0\times \quad (\text{linear in } Q \text{ for fixed } D, d)$$

## Appendix B — Dataset Class Reference

| Dataset | Class Index | Class Name    | Typical Object Size | Aerial Detection Difficulty         |
|---------|-------------|---------------|---------------------|-------------------------------------|
| AI-TOD  | 0           | airplane      | 15–40 px            | Moderate — distinct shape           |
| AI-TOD  | 1           | bridge        | 20–80 px            | Easy — elongated, distinctive       |
| AI-TOD  | 2           | storage-tank  | 8–20 px             | Hard — circular, low contrast       |
| AI-TOD  | 3           | ship          | 10–30 px            | Hard — elongated tiny profile       |
| AI-TOD  | 4           | swimming-pool | 6–16 px             | Very hard — small, similar to water |
| AI-TOD  | 5           | vehicle       | 4–12 px             | Critical — sub-8 px majority        |
| AI-TOD  | 6           | person        | 2–6 px              | Extreme — often sub-pixel           |
| AI-TOD  | 7           | wind-mill     | 4–10 px             | Very hard — thin, sparse            |

| Dataset  | Class Index | Class Name | Typical Object Size | Aerial Detection Difficulty      |
|----------|-------------|------------|---------------------|----------------------------------|
| VisDrone | 0           | pedestrian | 8–25 px             | Hard — tiny, occluded            |
| VisDrone | 1           | people     | 8–25 px             | Hard — group clustering          |
| VisDrone | 2           | bicycle    | 10–30 px            | Moderate — distinct shape        |
| VisDrone | 3           | car        | 20–60 px            | Moderate — primary class         |
| VisDrone | 4           | van        | 25–80 px            | Moderate — larger than car       |
| VisDrone | 5           | truck      | 30–100 px           | Easy — large, distinctive        |
| VisDrone | 6           | tricycle   | 10–30 px            | Moderate                         |
| VisDrone | 7           | awning-tri | 10–30 px            | Moderate                         |
| VisDrone | 8           | bus        | 40–120 px           | Easy — very large profile        |
| VisDrone | 9           | motor      | 8–20 px             | Hard — small, similar to bicycle |